Blood Donation Management System Software Requirement Specification (SRS)

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1. Introduction

1.1 Purpose

The purpose of this document is to provide a complete description of the functional and non-functional requirements for the **Online Blood Donation Management System**. This system aims to create an efficient, reliable, and user-friendly platform that connects blood donors with hospitals in urgent need of blood transfusions in Egypt.

The system will reduce the time and effort required for blood donation by allowing donors to register via a simple mobile/web interface and receive notifications when nearby hospitals need their specific blood type. It will also help hospitals find the nearest compatible donors quickly, minimize manual paperwork, and improve blood availability during emergencies, accidents, surgeries, pregnancy complications, and cancer treatments.

1.2 Scope

The proposed system includes:

- Web-based application for hospitals and administrators.
- Centralized MySQL database to store donor information, hospital details, and blood requests.
- Real-time notification system (using Firebase Cloud Messaging).
- Location-based search for nearest donors.
- Admin (customer service) module to manage users, monitor stock, and send confirmations/reminders.

Out of Scope:

- mobile application (can be added in future).
- Physical blood storage and lab testing management.
- Payment gateway for blood units.

1.3 Definitions, Acronyms, and Abbreviations

- **Donor**: General public who wishes to donate blood.
- **Hospital**: Authorized medical facility requesting blood.
- **Admin/Customer Service**: System administrator responsible for oversight.
- **FCM**: Firebase Cloud Messaging.
- **Haversine**: Formula used to calculate distance between two geographic points.

2. Overall Description

2.1 Product Perspective Currently, blood banks in Egypt are either independent or limited to specific hospital networks. This leads to delays in finding donors, especially for rare blood types, and forces hospitals to ask patients' families to donate or pay for blood. The proposed system provides a unified platform that connects donors directly with nearby hospitals in real time, making the process faster, safer, and more accessible.

2.2 Product Functions

- Donor registration and profile management.
- Hospital registration and blood request creation.
- Automatic matching and notification of nearest compatible donors.
- Admin management of users and requests.
- Post-donation confirmation and 3-month reminder.
- Information screen about blood donation rules.
- Smart Compatibility Engine: Utilizing a blood compatibility matrix to match donors and recipients beyond identical types.
- Automated Deferral Logic: System-enforced 3-month waiting period and medical rejection based on physiological screening results.

2.3 User Classes and Characteristics

1. **Blood Donors (General Public)**: Age 18-60, basic smartphone literacy, want simple and fast registration.
2. **Hospitals**: Authorized medical staff, need quick access to nearby donors.
3. **System Administrators (Customer Service)**: University/hospital staff, responsible for monitoring and support.

2.4 Operating Environment

- **Web**: React+ MySQL + node.js
- **Database**: MySQL.
- **Authentication**: firebase authentication.
- **Location**: GPS on donor's mobile.

2.5 Design and Implementation Constraints

- Use node.js + MySQL for backend.
- Passwords must be hashed (not stored in plain text).

2.6 Assumptions and Dependencies

- Users have internet access.
- Hospitals are pre-approved by admin.
- Donors provide accurate location and contact information.
- Google Maps/Geocoding API (optional for better location accuracy).

3. Specific Requirements

3.1 Functional Requirements

3.1.1 Donor Module (Web)

- FR-D1: Donor shall be able to register with name, age, phone, address, blood type, gender, and location, donation count, weight, Hight, last donation date, diagnosis, image.
- FR-D2: Donor shall login using username and password.
- FR-D3: Donor shall receive push notification when a nearby hospital needs his/her blood type.
- FR-D4: Donor shall be able to Accept or Reject a blood donation request.
- FR-D5: Donor shall view details of the blood request (hospital name, location, contact).
- FR-D6: Donor shall view information page about donation rules and eligibility.

3.2.2 Hospital Module (Web)

- FR-H1: Hospital shall register with name, phone, address, and location, license number.
- FR-H2: Hospital shall login.
- FR-H3: Hospital shall create a blood request (blood type needed, quantity, urgency).
- FR-H4: Hospital shall view list of nearby compatible donors with distance.
- FR-H5: Hospital shall contact donors directly via the displayed phone number.
- FR-H6: Hospital shall monitor and update the blood stock.
- FR-H7: Hospital shall approve or reject donors.

3.2.3 Admin Module (Web)

- FR-A2: Admin shall log in special credentials.
- FR-A3: Admin shall approve/reject hospital registrations.
- FR-A4: Admin shall monitor donors and hospitals.
- FR-A5: Admin shall monitor all blood requests and their status.
- FR-A6: Admin shall generate reports (number of donors, requests, blood types availability).
- FR-A7: Admin shall manage and reply to messages received from the Contact Us page.

3.2.4 System Features

- Automatic matching of blood type and location using Haversine formula.
- Real-time notification system for urgent requests.
- **Public informational pages**:
  - **About Us** page containing project information, vision, mission, team details, and objectives.
  - **Contact Us** page with a form allowing donors, hospitals, and visitors to send messages or inquiries directly to the administration.
- Security: Password fields are masked during registration and login.
- Responsive web design compatible with all screen sizes.
- The system shall send blood donation requests to multiple compatible donors simultaneously based on blood type, location, and availability.
- The search process shall continue until the hospital confirms that the blood need has been fulfilled.
- Accepting a request by one donor does not stop notifications to other donors until the request status is updated to “Fulfilled” by the hospital or admin.
- **Automated Confirmation:** The system shall automatically send a confirmation message/notification to the donor after the donation is marked as completed by the hospital or admin.
- **Automated Reminder:** The system shall automatically send a reminder notification to the donor 3 months after their last donation date to encourage future donations.

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

- System shall respond to requests within 5 seconds.
- Notification shall be delivered within 30 seconds of request creation.

3.3.2 Security Requirements

- All passwords shall be stored using strong hashing (bcrypt/PHP password_hash).
- Role-based access control (Donor / Hospital / Admin).
- Protection against SQL injection and XSS.
- "**Medical Data Privacy**: All clinical screening results and health history must be encrypted. Access is strictly restricted to authorized medical personnel to ensure patient confidentiality (HIPAA-compliant principles)."

3.3.3 Reliability and Availability

- System shall be available 24/7.
- Database backup shall be performed daily.

3.3.4 Usability

- Interface shall be simple and easy to use for all age groups.
- system shall have intuitive navigation.
- The web application shall be fully responsive, providing an optimal viewing and interaction experience across a wide range of devices, including mobile phones, tablets, and desktop computers.

3.3.5 Scalability

- System shall support adding mobile app and cloud hosting in future.